H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME: **Mark Scheme**

CLASS:

BIOLOGY

9648/03

Paper 3 Applications and Planning

September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	**60**
For Examiner's Use	
4	**12**

This document consists of **15** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 Bacteria can be genetically modified to produce insulin for human use. To achieve this, human insulin genes are transferred into bacteria. Plasmids containing two antibiotic resistance genes, one coding for resistance to tetracycline and one for resistance to ampicillin, are used to carry out this transfer.

A restriction enzyme was used to cut up the human DNA and plasmids. **Fig. 1.1** shows the different fragments of human DNA and the type of cut plasmid that was produced.

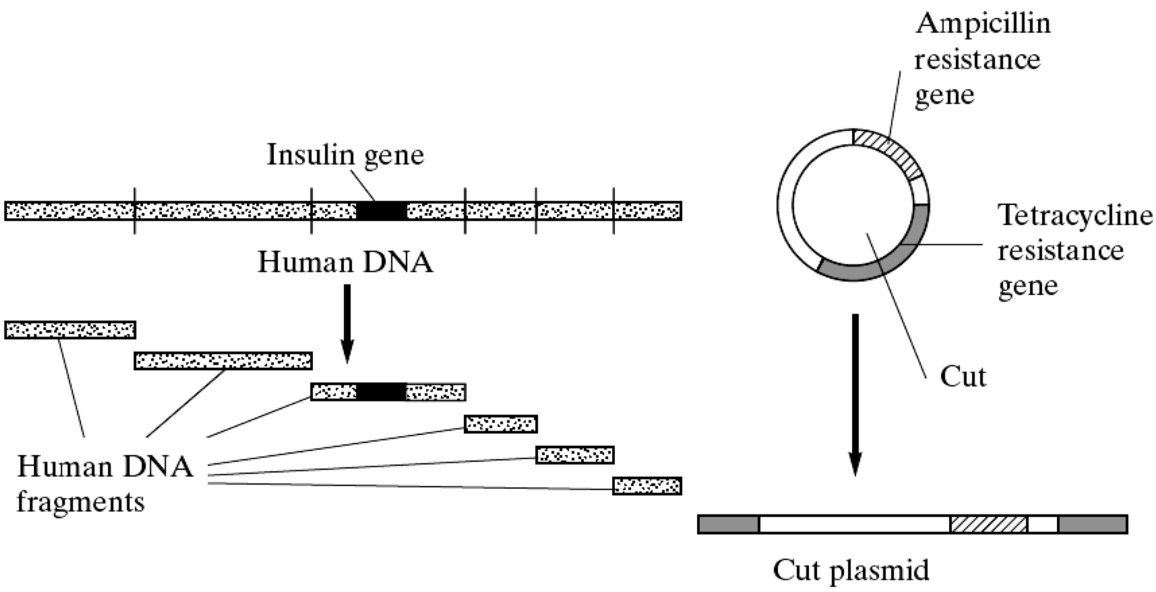


Fig. 1.1

(a) Describe a plasmid.

circular double stranded DNA;

separate from main bacterial DNA;

contains only a few genes;

replicates autonomously / independently of bacteria DNA;

[2]

(b) Suggest why the restriction enzyme has cut the human DNA in many places but has cut the plasmid DNA only once.

enzymes only cut DNA at specific base sequence / recognition site / specific restriction site;

sequence of bases / recognition site / specific restriction site (on which enzyme acts) occurs once in plasmid and many times in human DNA;

[2]

The fragments of human DNA and the cut plasmids were mixed together with DNA ligase. Several types of plasmid were formed. Some contained human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmid are shown in **Fig 1.2**.

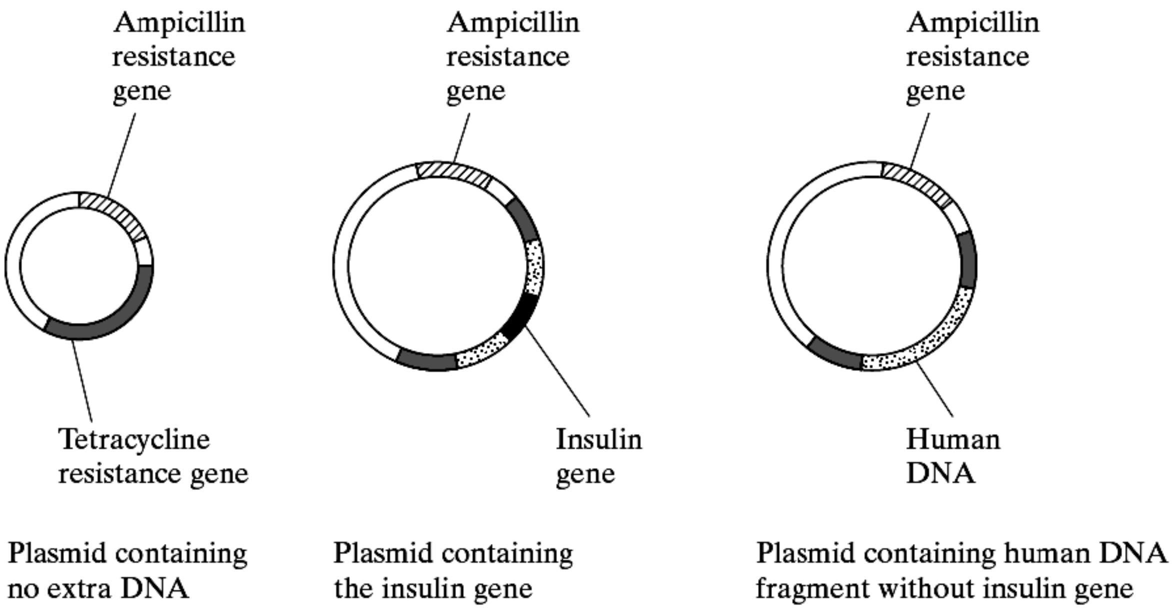


Fig 1.2

(c) Explain what causes several types of plasmid to be formed.

all cut DNA have same / complementary base sequence sticky ends which anneals by complementary base pairing (between A=T and C≡G);

It is a random process by which sticky ends join and subsequently ligated by DNA ligase;

[2]

(d) Outline the reaction that DNA ligase catalyses.

Formation of phosphodiester bonds between nucleotides (by condensation reaction with the removal of water);

[1]

The plasmids are mixed with the bacteria. Some bacteria take up the plasmids.

Some of the plasmids did not take up the extra human DNA. Replica plating was used to identify the bacteria with the human DNA. **Fig. 1.3** shows the bacterial colonies that grew on two replica plates.

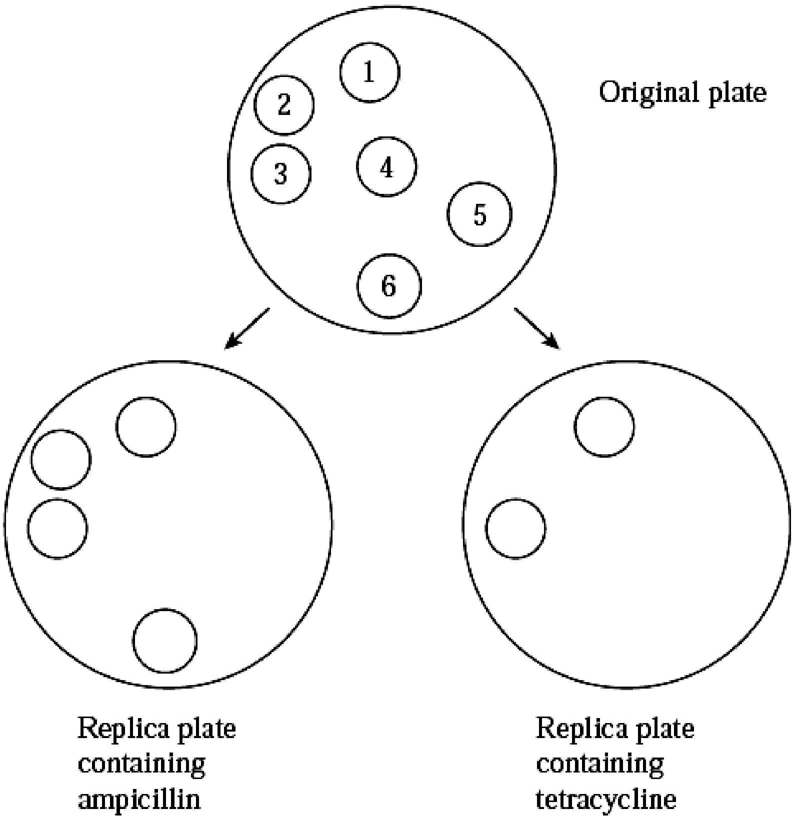


Fig. 1.3

(e) With reference to Fig. 1.3, explain the results of the replica plate containing:

(i) ampicillin.

Colonies 1,2 3 & 6 have taken up the plasmid;

Because they are resistant to ampicillin / has ampicillin resistance gene;

OR

Colonies 4 & 5 did not take up the plasmid;

Because they are not resistant to ampicillin / lack of ampicillin resistance gene;

[2]

(ii) tetracycline.

Colonies 1 & 3 do not have the recombinant plasmid;

Because they are still resistant to tetracycline / has tetracycline resistance gene;

Colonies 2 and 6 dies because taken up the plasmid with human DNA / recombinant plasmid;

Tetracycline resistance gene disrupted / insertional inactivation of tetracycline resistance gene;

[3]

(f) Describe how the bacteria containing the insulin gene are used to obtain sufficient insulin for commercial use.

use of fermenters;

provides nutrients plus suitable conditions for optimum growth / named environmental factor;

reproduction of bacteria via binary fission increasing the expression of insulin gene / more insulin proteins are produced;

insulin accumulates and is extracted and <u>purified</u>;

[3]

[Total: 15]

2 Cord blood is the blood that circulates through the umbilical cord from the foetus to the placenta. The umbilical cord that remains attached to the placenta has been found to be rich in cord blood stem cells. Cord blood stem cells are the young or immature cells that can transform into other forms of essential blood cell types, such as red blood cells, white blood cells and platelets.

Adapted from : http://www.cordlife.com/sg/en/umbilical-cord-tissue/about-umbilical-cord-tissue

(a) Describe the unique features of all stem cells.

Stem cells are unspecialised, there is an absence of tissue-specific structures for specialised functions;

Stem cells are capable of dividing by mitosis and continually renewing themselves for long periods;

Stem cells are capable of differentiating into specialised cell types under appropriate conditions;

[3]

(b) Suggest an advantage of using cord blood stem cells in place of blood stem cells from the bone marrow.

It is not painful as blood is drawn from umbilical vein after umbilical cord is separated from the newborn baby / no surgical procedure involved;

Cord blood stem cells are easier to collect as blood is drawn from umbilical vein after umbilical cord is separated from the newborn baby / no surgical procedure involved;

There is no risk to the baby or mother as blood is drawn from umbilical vein after umbilical cord is separated from the newborn baby / no surgical procedure involved;

[1]

(c) State the potency of cord blood stem cells.

Multipotent;

[1]

There exist other groups of stem cells of greater developmental potential.

(d) Describe, with a named example, a group of stem cells with higher developmental potential than cord blood stem cells.

Zygotic stem cells are totipotent;

These cells can differentiate into any cell types to form whole organisms;

Embryonic stem cells are pluripotent;

able to give rise to almost all cell types to form the three primary germ layers; ectoderm, endoderm and mesoderm / to the multiple specialised cell types / tissues / organs in the developing foetus;

[2]

Besides using stem cell transplant to treat genetic disorders such as SCID, gene therapy could be another treatment option. Severe Combined Immunodeficiency (SCID) is a disease of young children in which patients have neither cell-mediated immune responses nor the ability to make antibodies.

There are two types of SCID: Adenosine Deaminase (ADA) deficient SCID and X-linked SCID. The delivery system used in SCID gene therapy is usually viral gene delivery system.

(e) Suggest a type of virus that can be used in SCID gene therapy.

Retrovirus;

[1]

(f) Explain the factors that keep gene therapy from becoming an effective treatment for SCID.

Gene therapy is <u>short-lived</u> and requires <u>frequent</u> / <u>repeated</u> treatments;

Incorrect insertion of normal gene into tumour-suppressor gene may cause cancer / insertional mutagenesis;

Immune response may be triggered as a foreign viral vector is introduced;

Problem with controlling the activity of gene expression;

[3]

[Total: 11]

3 Golden rice is a transgenic variety of rice that has been genetically engineered to be rich in beta-carotene (a pre-cursor to vitamin A). **Fig. 3.1** below shows an artificial DNA sequence used in the production of golden rice.

Key:

pro – promoter sequence

ter – terminator sequence

Hyg resist – antibiotic resistance from a bacterium

Fig. 3.1

(a) Explain what is meant by *transgenic*.

Contains DNA from two different species rice and daffodil

OR

Rice which contains a foreign DNA from another species daffodil;

[1]

(b) State the organism from which the promoter sequence for *Hyg* resist were taken from and explain your choice.

Rice / plant;

So that can be recognized by (eukaryotic) RNA polymerase found in rice to be expressed;

[2]

(c) Briefly outline one way in which this gene construct may be inserted into rice cells.

Inserted into (Ti) plasmid / formation of recombinant plasmid in *Agrobacterium*;

Agrobacterium infects rice cell to incorporate gene construct into rice cell;

OR

DNA containing gene coated on to gold particles;

Shot with gene gun / biolistics directly into rice cell;

[2]

(d) State why it is not possible to produce golden rice by breeding.

Rice and daffodils belong to different species which will not naturally interbreed to produce fertile viable offspring;

Even if interbreeding occurred, it involves random fusion of gametes which will not necessarily contain only favourable traits.

[1]

Hyg resist confers resistance to hygromycin, an antibiotic which kills both bacterial cells and plant cells.

(e) Explain the role of the *Hyg* resistance gene in this procedure.

It is a selectable marker to enable the selection of the transformed / transgenic cells;

only the cells with the new DNA can grow in the presence of the antibiotic;

[2]

(f) A concern of having *Hyg* resistance in the golden rice is the possible horizontal transfer of *Hyg* resist to other organisms.

(i) Suggest how *Hyg* resistance can be acquired by other plants and state a technical safeguard that can minimize this risk.

Transgenic plants may pass their new genes (for hyg-resistance) to close relatives in nearby wild areas through pollen transfer;

Greater isolation distances between the transgenic crop and nearby areas (fields, forests etc);

OR

Planting a border of unrelated plants with which transgenic plants would not hybridize through spread of transgenic pollen;

[2]

(ii) *"Hyg* resistance may result in a resistant strain of bacteria for which this antibiotic will no longer be effective." Explain if this claim is biologically valid.

Yes; When humans ingest golden rice, the resistance gene may be transferred to gut bacteria;

No; The resistance gene will probably be broken down by the time it reaches the gut.

No; Direct transfer between plants and bacteria is unlikely since they are highly unrelated species.

[2]

(g) Suggest 2 possible benefits of golden rice.

May help to prevent night blindness;

Reduce vitamin A deficiency in developing countries;

Enable local farmers to grow a cash crop;

Enable the development of rice breeding to improve local crops;

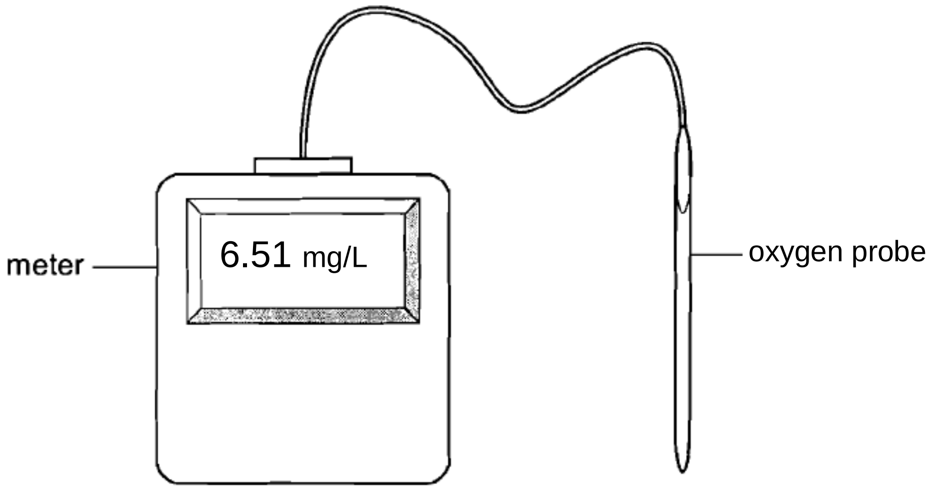
[2]

[Total: 14]

Planning Question

4 You are required to plan, but not carry out, an investigation into the effect of temperature on rate of photosynthesis.

Fig. 4.1 shows an oxygen probe and meter that can be used to measure the concentration of dissolved oxygen in a solution in mg/L.



Design an experiment, using an oxygen sensor to test the hypothesis that:

The rate of photosynthesis is dependent on the temperature.

Your planning can include, but is not limited to the following equipment & materials.

- A beaker of suspension of unicellular algae in water
- Bench lamp
- thermometer
- hot water at 80^{o}C
- stopwatch
- distilled water
- glass rod
- Stopwatch
- a variety of different sized beakers, test tubes, measuring cylinders, syringes, droppers and pipettes for measuring volumes

Your plan should include:

- an explanation of the theory to support your practical procedure,
- relevant, clearly labelled diagram(s) to show the arrangement of apparatus used,
- an explanation of dependent & independent variables involved,
- proposed layout of results tables and graphs with clear headings and labels,
- full details and explanations of the procedures you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible,
- correct use of scientific & technical terms

[Total: 12]

Theoretical Background (2 m max)

Temperature is a limiting factor of photosynthesis;

Temperature affects rate of enzymatic reactions during light dependent reaction (e,g, NADP reducatase) and light independent reaction (e.g, Rubisco)

Oxygen is released during photolysis of water and can be used to determine rate of photosynthesis.

Variables

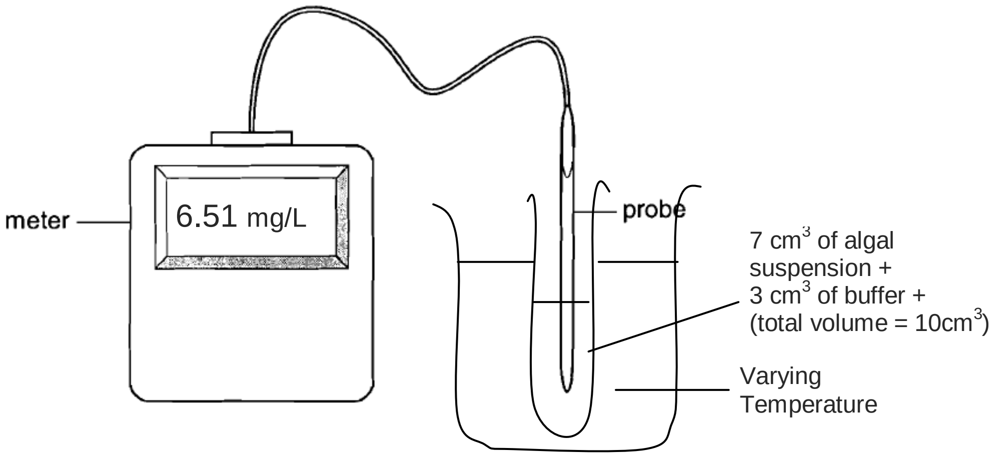
Independent variable: Temperature

Dependent variable: Concentration of dissolved oxygen

Variable kept constant:

Volume of algae suspension, wavelength of light, light intensity, pH

Hypothesis (same as conclusion): The rate of photosynthesis is maximum at optimum temperature and decreases away from optimum temperature.



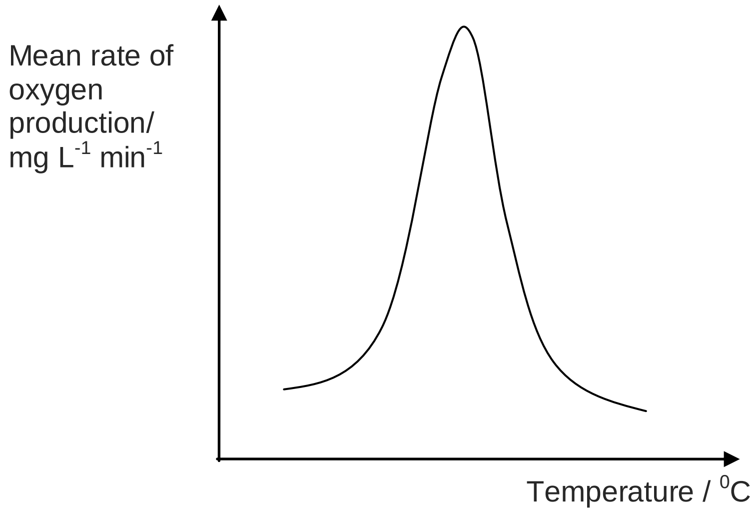
Procedures

1. Using a beaker of hot water and thermometer, prepare a water bath fixed at 30^{0}C.
2. Label 5 boiling tubes A – E.
3. Using a syringe, add 7.0 cm^3 of algae suspension and 3.0 cm^3 of pH buffer.
4. Place the boiling tube into the water bath and allow for 5 minutes of equilibration.
5. Using a ruler, place the bench lamp 10 cm away from the water bath.
6. Switch on the bench lamp.
7. After 5 min, insert the oxygen probe into the boiling tube and ensure solution covers probe. Record the concentration of dissolved oxygen detected.
8. Wash/ rinse the probe with distilled water and put it aside, ready for the next reading.
9. Repeat steps 3 to 8 for the 4 other temperatures of 40^{0}C, 50^{0}C, 60^{0}C and 70^{0}C.
10. Repeat steps 3 to 9 twice and calculate the mean concentration of dissolved oxygen.
11. Set up the control by repeat the experiment steps 3 to 8 but carrying out in the dark.
12. Record the data in the table below and calculate rate of reaction as 1/time taken. Calculate the mean rate of oxygen produced per minute by dividing by 5 minutes.

Table 1: Table of results of rate of reaction

Temperature/ ^{0}C	**Concentration of dissolved oxygen in 5 minutes/ mg L^{-1}**				**Mean rate of oxygen production/ mgL^{-1} min^{-1}**
	1	**2**	**3**	**Mean**	
30					
40					
50					
60					
70					

13. Plot the graph of mean rate of oxygen production / mgL^{-1} min^{-1} against temperature / ^{0}C.



Conclusion

The rate of photosynthesis is maximum at optimum temperature and decreases away from optimum temperature.

Risk Assessment

1. Scalding by hot water. Use cloth to wrap the mouth of beaker.
2. Electrocution due to thermostat water bath. Avoid touching electrical appliance with water.

Mark scheme

Theoretical background [2m max]

T1: Temperature is a limiting factor of photosynthesis;

T2: Temperature affects rate of enzymatic reactions during light dependent reaction (e,g, NADP reducatase) and light independent reaction (e.g, Rubisco)

T3: Oxygen is released during photolysis of water and can be used to determine rate of photosynthesis.

Procedures [8m max]

CV1: Fixed volume of algae suspension / volume of buffer / Light intensity / wavelength of light / total volume kept constant;

D: Diagram of experimental set-up (with water bath); ® 80^{O}C

E: Equilibration time

C: Control by replacing conducting experiment in the dark;

R: Repeat experiment three times + calculate mean time;

PD: Calculate mean rate of oxygen production by dividing by time;

TB: tabulation of data with correct headings and units;

G: correct graph with axes labels & units + trend;

CON: optimum temperature;

DV: Dependent variable: concentration of dissolved oxygen + independent variable of at least 5 temperatures.

Risk assessment [2m max]

RA1: Scalding by hot water. Use cloth to wrap the mouth of beaker. ® glove

RA2: Electrocution due to thermostat water bath. Avoid touching electrical appliance with water.

RA3: Putting thermometer back into basket/ put away safely to prevent breakage because the mercury fumes released when the thermometer is broken is toxic.

Free Response Question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Using your knowledge of the genetic basis of cystic fibrosis, compare liposomal and viral delivery systems in the treatment of cystic fibrosis using gene therapy. [6]

(b) Describe how RFLP analysis can be used in genome mapping. [8]

(c) Discuss the ethical concerns that have arisen regarding the Human Genome Project. [6]

[Total: 20]

(a) Scientist use either viral (e.g. adenovirus) or non-viral delivery (liposomes) methods to introduce a normal functional CFTR allele into the human.

	Point of comparison	Adenovirus-mediated gene therapy	Liposome-mediated gene therapy
	Differences		
D1	Immune response	Adenovirus more likely to evoke immune response	Less likely to evoke immune response
D2	Effectiveness on evoking immune response	Less likely to be effective after the body has developed immunity to the viral vector	Remains effective as body does not develop immunity to the vector
D3	Disease causing / risk assessment / Toxicity	Adenoviral vector (which does not integrate into the genome can cause throat infections but not human malignancies (cancer)	Do not cause disease at all / safer (Liposomes have a small degree of toxicity – so many rounds of treatment can be harmful to the cells or tissues being treated)
		(Idea that viral delivery methods cause more harm or health implications than the liposomal method)	
D4	Specificity	More specific targeting of cells	Less specific targeting of cells
D5	Transfection efficiency	High transfection efficiency.	Low transfection efficiency.
		(Idea that viral delivery methods allow functional gene to be more easily delivered into the host cell than the liposomal method)	
D6	Delivery of normal functional CFTR allele into nucleus / Chances of integration into the genome	Greater chances of delivering normal functional allele into nucleus via nuclear pore as adenovirus directly injects DNA at nuclear pore.	No ability to transfer normal functional allele into nucleus.

		(Idea that viral delivery methods allow higher success of functional gene integrating with host cell genome as compared to the liposomal method)	
D7	Entry into cell	Virus enters cell via endocytosis, delivering normal functional CFTR allele into target cell.	Fusion of lipid bilayer of liposome with cell membrane leads to delivery of normal functional CFTR allele into cell.
	Similarities		
S1	Integration	Liposomes and adenoviruses not able to integrate normal functional CFTR allele into genome (extrachromosomal) / do not cause insertional mutagenesis; And hence the treatment is short-lived (due to the transient nature of the gene that has been inserted into the cell cytoplasm);	
S2	Duration of therapeutic effects	Treatment is transient as epithelial cells are constantly being shed, so repeated therapy is needed;	
S3	Target organs	Cystic fibrosis affects other organs that are less accessible e.g., pancreas and small intestines, and the epithelial cells of the pancreas is a challenge to target by both liposomal or viral delivery; *(Idea that it is difficult to target all the affected organs)*	
S4	Target cells	Difficult to target the **stem cells** of the epithelium of every organ *(Idea that the inserted gene is not inheritable because the gene is only delivered to somatic cells)*	
S5	*In vivo / ex vivo* approach	Both systems are carried out using an *in vivo* approach because it is difficult to extract epithelial cells from respiratory tract.	
S6	Level of expression	Fine-tuning the expression of normal functional CFTR allele within target cells is difficult; under-expression of normal functional CFTR allele in target cells would render gene therapy ineffective.	

(b)

1 RFLP is a phenomenon whereby various **lengths of restriction fragments** are generated from DNA sequences using **restriction enzymes**

+ arises due to **different nucleotide sequences** or / **variable number of tandem repeats** (VNTR);

2 RFLP analysis: involves amplification by **PCR** → cutting by **restriction enzymes** → separation by **gel electrophoresis** → transfer by **Southern blotting** **+ nucleic acid hybridization** with radioactive probe;

3 Definition of **RFLP marker**: a segment of DNA found at a specific site along a chromosome and can be uniquely recognized by probes;

+ RFLP markers are **inherited in a Mendelian fashion**;

4 RFLP analysis can be performed on **RFLP markers located on two different loci** of the genome

+ **2 individuals**, each being **homozygous for each RFLP marker**, but have **different band patterns** for each particular RFLP marker, can be crossed to **produce heterozygous offsprings**;

5 **Heterozygotes** are then **crossed with one of the homozygous parent**

+ offspring can **inherit pairs of RFLP markers like the parents** or **inherit recombinant pairs** of RFLP markers;

6 **If** the 2 gene loci are **not linked**, number of offspring showing the four phenotypes will be of ratio **1:1:1:1**;

7 If the 2 gene loci are **linked without crossing over**, there will only be **2 phenotypes** observed

+ as the **two gene loci will assort and segregate together**;

8 If the two gene loci are **closely linked**, the **proportion of offspring with recombinant phenotypes will be less than parental phenotypes**;

9 If the two gene loci are linked, frequency at which genetic markers are inherited together with gene loci, can be used to **calculate distance between both gene loci** based on frequency of crossing-over;

10 By performing **analysis on many other pairs of RFLP markers** and with other cytogenetic studies, the researcher is able to come up with a detailed map of the genome;

(c)

1 Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others / Discrimination by institutions ;

2 Privacy and confidentiality of genetic information;

3 Psychological impact and stigmatization due to an individual's genetic differences / Discrimination by individuals;

4 Reproductive issues including adequate informed consent for complex and potentially controversial procedures / use of genetic information in reproductive decision making / reproductive rights;

5 Clinical issues including the education of doctors and other health service providers, patients, and the general public in genetic capabilities, scientific limitations, and social risks / implementation of standards and quality-control measures in testing procedures;

6 Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions;

7 Conceptual and philosophical implications regarding human responsibility, free-will versus genetic determinism / concepts of health and disease / Religious concerns, doctrinal issues ;

8 Health and environmental issues concerning genetically modified foods (GM) and microbes;

9 Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials;